SELECTIVE SEPARABILITY AND ITS VARIATIONS

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Abstract. A space X is said to be selectively separable (=M-separable) if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that $\bigcup\{F_n : n \in \omega\}$ is dense in X. On selective separability and its variations, we show the following: (1) Selective separability, R-separability and GN-separability are preserved under finite unions; (2) Assuming CH (the continuum hypothesis), there is a countable regular maximal R-separable space X such that X^2 is not selectively separable; (3) $\{0,1\}^{\mathfrak{c}}$ has a selectively separable, countable and dense subset S such that the group generated by S is not selectively separable. These answer some questions posed in [6].

1. INTRODUCTION

All spaces are assumed to be regular. The symbol $\mathbb{D}$ is the discrete space consisting of the two points 0 and 1. For a Tychonoff space X, we denote by $C_p(X)$ (resp., $C_p(X, \mathbb{D})$) the space of all real-valued (resp., $\mathbb{D}$-valued) continuous functions on X with the topology of pointwise convergence. Unexplained notions and terminology are the same as in [10].

Definition 1.1 ([5], [6])**.** A space X is *selectively separable* (=*M-separable*) if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that $\bigcup\{F_n : n \in \omega\}$ is dense in X. A space X is *R-separable* if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are points $x_n \in D_n$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is dense in X. A space X is *H-separable* if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that every nonempty open set in X meets all but finitely many F_n. A space X is *GN-separable* if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are points $x_n \in D_n$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is *groupable*, i.e., $\{x_n : n \in \omega\}$ can be partitioned into nonempty finite sets F_n $(n \in \omega)$ such that every nonempty open set in X meets all but finitely many F_n.

Selective separability (resp., GN-separability) was first introduced and studied in Scheepers [22] (resp., Di Maio et al. [15]).

A family $\mathcal{B}$ of nonempty open sets of a space X is said to be a *π-base* if every nonempty open set of X contains some member of $\mathcal{B}$. Obviously every space with a countable π-base is R-separable and H-separable, every R-separable (or H-separable) space is selectively separable, and every GN-separable space is dense-in-itself (i.e., there is no isolated point). Juhász and Shelah [13] showed that if

2000 *Mathematics Subject Classification.* 54C35; 54D65; 54E65.

Key words and phrases. selectively separable, M-separable, R-separable, H-separable, GN-separable, Menger, maximal.

The second author was supported by KAKENHI (No. 22540154).

X is compact and every dense subset of X is separable, then X has a countable π-base. Hence among compact spaces, selective separability and the existence of some countable π-base are equivalent.

We recall some facts on selective separability. A space X is said to have *countable fan-tightness* [1] if whenever $A_n \subset X$ and $x \in \overline{A}_n$ $(n \in \omega)$, there are finite sets $F_n \subset A_n$ $(n \in \omega)$ such that $x \in \overline{\bigcup\{F_n : n \in \omega\}}$.

Lemma 1.2 ([6], Propositions 2.2, 2.3)**.** *The following hold.*

(1) *selective separability is hereditary with respect to open subspaces and dense subspaces,*
(2) *every space having a selectively separable, open and dense subset is selectively separable,*
(3) *every separable space with countable fan-tightness is selectively separable.*

2. Finite unions

Bella et al. asked in [6, Problem 3.8] whether selective separability is preserved under finite unions. We show that the answer is affirmative.

Lemma 2.1. *A space X is selectively separable if and only if every decreasing sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that $\bigcup\{F_n : n \in \omega\}$ is dense in X.*

Proof. Let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X. For each $n \in \omega$, we put $E_n = \bigcup_{k \geq n} E_k$. We take finite sets $F_n \subset E_n$ $(n \in \omega)$ such that $\bigcup\{F_n : n \in \omega\}$ is dense in X. For each $n \in \omega$, we put $F_n = F_{n,n} \cup F_{n,n+1} \cup \cdots \cup F_{n,k_n}$, where $k_n \geq n$ and $F_{n,j} \subset D_j$ for $n \leq j \leq k_n$. For each $n \in \omega$, let $G_n = F_{0,n} \cup F_{1,n} \cup \cdots \cup F_{n,n}$. Then $G_n \subset D_n$ and $\bigcup\{G_n : n \in \omega\}$ is dense in X. □

Theorem 2.2. *The finite union of selectively separable spaces is selectively separable.*

Proof. It suffices to prove that if $X = A \cup B$, where A and B are selectively separable, then X is selectively separable. Let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X. In view of Lemma 2.1, we may assume $D_{n+1} \subset D_n$ $(n \in \omega)$. For each $n \in \omega$, we put $U_n = X \setminus \overline{A \cap D_n}$. Then $\{U_n : n \in \omega\}$ is an increasing family of open sets in X. For each $n \in \omega$, note that $U_n \cap D_k$ $(k \geq n)$ are dense in U_n, $U_n \cap D_k \subset U_n \cap B \subset U_n$, and $U_n \cap B$ is selectively separable by Lemma 1.2 (1). Hence there are finite sets $F_{n,k} \subset U_n \cap D_k$ $(k \geq n)$ such that $\bigcup\{F_{n,k} : k \geq n\}$ is dense in U_n. For each $n \in \omega$, we put $F_n = \bigcup\{F_{k,n} : k \leq n\}$. Then F_n is a finite subset of D_n and $\bigcup\{F_n : n \in \omega\}$ is dense in $\bigcup\{U_n : n \in \omega\}$. Let V be the interior of the set $\bigcap\{\overline{A \cap D_n} : n \in \omega\}$. For each $n \in \omega$, note that $V \cap A \cap D_n$ is dense in V, $V \cap A \cap D_n \subset V \cap A \subset V$, and $V \cap A$ is selectively separable by Lemma 1.2 (1). Hence there are finite sets $G_n \subset V \cap A \cap D_n$ $(n \in \omega)$ such that $\bigcup\{G_n : n \in \omega\}$ is dense in V. Then $F_n \cup G_n$ is a finite subset of D_n and $\bigcup\{F_n \cup G_n : n \in \omega\}$ is dense in X. □

Theorem 2.3. *The finite union of R-separable (resp., GN-separable) spaces is R-separable (resp., GN-separable).*

Proof. Let $X = A \cup B$, where A and B are R-separable (resp., GN-separable), and let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X. We need to show that there are $d_n \in D_n$ $(n \in \omega)$ such that $\{d_n : n \in \omega\}$ is dense (resp., groupable) in X.

Let A' be a countable dense subset of A. For each $a \in A'$, let $J_a = \{n \in \omega : a \notin \overline{D_n \cap A}\}$. Let $A'' = \{a \in A' : |J_a| < \omega\}$. Let $U = \mathrm{Int}_X(\overline{A''})$ and $V = X \setminus \overline{A''}$. Of course, $U \cup V$ is dense open in X.

Now $\{J_a : a \in A' \setminus A''\}$ is a countable collection of infinite subsets of ω, so there is an infinite pairwise-disjoint collection $\mathcal{W}$ of infinite subsets of ω such that $W \in \mathcal{W}$ and $a \in A' \setminus A''$ imply $W \cap J_a \neq \emptyset$.

Claim. If $W \in \mathcal{W}$, then:

(1) $\bigcup\{D_i \cap A \cap U : i \in W\}$ *is dense in* U*;*
(2) $\bigcup\{D_i \cap B \cap V : i \in W\}$ *is dense in* V*.*

If $a \in A''$, then J_a is finite, so $a \in \overline{D_i \cap A}$ for all but finitely many $i \in W$. Since $A'' \cap U$ is dense in U, (1) follows.

To see (2), suppose otherwise, and let O be a nonempty open subset of V such that $O \cap D_i \cap B = \emptyset$ for each $i \in W$. Since $X = A \cup B$ and D_i is dense in X, it easily follows that $O \subset \overline{D_i \cap A}$ for each $i \in W$. Since A' is dense in A, there is some $a \in A' \cap O$. Then $a \in A' \setminus A''$, so there is some $i \in W \cap J_a$. Thus $a \in O \setminus \overline{D_i \cap A}$, contradiction.

Index $\mathcal{W}$ as W_n, $n \in \mathbb{Z}$, where $\mathbb{Z}$ is the set of integers. R-separability and GN-separability are inherited by open subspaces, $A \cap U$ and $B \cap V$ are R-separable (resp., GN-separable). So it follows from the Claim that there are $x_n \in \bigcup\{D_i \cap A \cap U : i \in W_n\}$ $(n \geq 0)$ such that $\{x_n : n \geq 0\}$ is dense (resp., groupable) in $U \cap A$ and hence in U, and $x_n \in \bigcup\{D_i \cap B \cap V : i \in W_n\}$ $(n < 0)$ such that $\{x_n : n < 0\}$ is dense (resp., groupable) in $V \cap B$ and hence in V. Then $\{x_n : n \in \mathbb{Z}\}$ is dense (resp., groupable) in X. Since $\mathcal{W}$ is a pairwise-disjoint collection, $\{x_n : n \in \mathbb{Z}\}$ is a selection of at most one element from each D_i. Hence X is R-separable (resp., GN-separable). □

Lemma 2.4. *If a space X has a countable open cover consisting of selectively separable (resp., R-separable, GN-separable) subsets, then it is selectively separable (resp., R-separable, GN-separable).*

Proof. Let $\{U_n : n \in \omega\}$ be an open cover consisting of selectively separable (resp., R-separable, GN-separable) subsets. For each $n \in \omega$, let $V_n = U_n \setminus \overline{\bigcup\{U_k : k \leq n-1\}}$. Then $\{V_n : n \in \omega\}$ is a pairwise-disjoint family of selectively separable (resp., R-separable, GN-separable) open subsets. Hence $V = \bigcup\{V_n : n \in \omega\}$ is also selectively separable (resp., R-separable, GN-separable). Since V is dense in X, X is selectively separable (resp., R-separable, GN-separable). □

Corollary 2.5. *If a space X is separable and has a locally finite cover consisting of selectively separable (resp., R-separable, GN-separable) subsets of X, then X is selectively separable (resp., R-separable, GN-separable).*

Proof. Let $\mathcal{C}$ be a locally finite cover consisting of selectively separable (resp., R-separable, GN-separable) subsets. Let $\{x_n : n \in \omega\}$ be a dense subset of X. Note that each x_n has an open neighborhood U_n which is contained in the union of finitely many members of $\mathcal{C}$. By Theorem 2.2, each U_n is selectively separable (resp., R-separable, GN-separable). By Lemma 2.4, $U = \bigcup\{U_n : n \in \omega\}$ is selectively separable (resp., R-separable, GN-separable). Since U is dense in X, X is selectively separable (resp., R-separable, GN-separable). □

In view of Lemma 2.1, it is natural to introduce the following notions.

Definition 2.6. A space X is *wR-separable* if for each decreasing sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are points $x_n \in D_n$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is dense in X. A space X is *wH-separable* if for each decreasing sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that every nonempty open set in X meets all but finitely many F_n. A space X is *wGN-separable* if for each decreasing sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are points $x_n \in D_n$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is groupable.

Note that every wGN-separable space is dense-in-itself. R-separability (resp., H-separability, GN-separability) implies wR-separability (resp., wH-separability, wGN-separability).

Lemma 2.7. *Let X be a space.*

(1) *X is GN-separable if and only if it is R-separable, wH-separable and dense-in-itself,*
(2) *X is wGN-separable if and only if it is wR-separable, wH-separable and dense-in-itself.*

Proof. (1). Let X be a GN-separable space. Then it is obviously R-separable. We observe that X is wH-separable. Let $\{D_n : n \in \omega\}$ be a decreasing sequence of dense subsets of X. By GN-separability of X, there are points $x_n \in D_n$ $(n \in \omega)$ and a partition $\{F_n : n \in \omega\}$ of the set $\{x_n : n \in \omega\}$ such that each F_n is a nonempty finite set and every nonempty open set in X meets all but finitely many F_n. Since $\{D_n : n \in \omega\}$ is decreasing, for each $n \in \omega$ $D_n \supset F_k$ holds for all but finitely many $k \in \omega$. Hence we can take a strictly increasing sequence $\{k_n : n \in \omega\} \subset \omega$ such that $D_n \supset F_{k_n}$ $(n \in \omega)$. This sequence $\{F_{k_n} : n \in \omega\}$ satisfies that every nonempty open set in X meets all but finitely many F_{k_n}. Conversely assume that X is R-separable, wH-separable and dense-in-itself. Let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X. By R-separability of X, there are points $x_n \in D_n$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is dense in X. Since X is dense-in-itself, each D_n is infinite, so we may assume $x_n \neq x_m$ whenever $n \neq m$. For each $n \in \omega$, let $E_n = \{x_k : k \geq n\}$. Since X is dense-in-itself, each E_n is dense in X. By wH-separability of X, there are nonempty finite sets $F_n \subset E_n$ $(n \in \omega)$ such that every nonempty open set in X meets all but finitely many F_n. Since each F_n is finite and $\{E_n : n \in \omega\}$ is decreasing, we can take a strictly increasing sequence $\{k_n : n \in \omega\} \subset \omega$ such that $\{F_{k_n} : n \in \omega\}$ is pairwise-disjoint. Take any partition $\{G_{k_n} : n \in \omega\}$ of $\{x_n : n \in \omega\}$ such that $F_{k_n} \subset G_{k_n}$ $(n \in \omega)$. Then every nonempty open set in X meets all but finitely many G_{k_n}.

(2) can be proved similarly. □

Theorem 2.8. *The finite union of wR-separable (resp., wH-separable) spaces is wR-separable (resp., wH-separable). In particular the finite union of wGN-separable spaces is also wGN-separable.*

Proof. The case of wR-separability can be proved more simply than the case of R-separability. Let $X = A \cup B$, where A and B are wR-separable. We show that X is wR-separable. Let $\{D_n : n \in \omega\}$ be a decreasing sequence of dense subsets of X. Let $\{J_n : n \in \omega\}$ be any partition of ω of infinite subsets such that the sequence $\{k_n = \min J_n : n \in \omega\}$ is increasing. For each $n \in \omega$, let $U_n = X \setminus \overline{A \cap D_{k_n}}$. For each $n \in \omega$, note that $U_n \cap D_k$ $(k \in J_n)$ is dense in U_n, $U_n \cap D_k \subset U_n \cap B \subset U_n$, and $U_n \cap B$ is wR-separable. Hence there are points $x_k \in U_n \cap D_k$ $(k \in J_n \setminus \{k_n\})$

such that $\{x_k : k \in J_n \setminus \{k_n\}\}$ is dense in U_n. Then $\{x_n : n \in \omega \setminus \{k_n : n \in \omega\}\}$ is dense in $\bigcup\{U_n : n \in \omega\}$. Let V be the interior of $\bigcap\{\overline{A \cap D_{k_n}} : n \in \omega\}$. For each $n \in \omega$, note that $V \cap A \cap D_{k_n}$ is dense in V, $V \cap A \cap D_{k_n} \subset V \cap A \subset V$, and $V \cap A$ is wR-separable. Hence there are points $x_{k_n} \in V \cap A \cap D_{k_n}$ $(n \in \omega)$ such that $\{x_{k_n} : n \in \omega\}$ is dense in V. Consequently $\{x_n : n \in \omega\}$ is dense in X.

Let $X = A \cup B$, where A and B are wH-separable. We show that X is wH-separable. Let $\{D_n : n \in \omega\}$ be a decreasing sequence of dense subsets of X. For each $n \in \omega$, let $U_n = X \setminus \overline{A \cap D_n}$. Without loss of generality, we may assume $U_n \neq \emptyset$ for all $n \in \omega$. For each $n \in \omega$, note that $U_n \cap D_k$ is dense in U_n $(k \geq n)$, $U_n \cap D_k \subset U_n \cap B \subset U_n$ and $U_n \cap B$ is wH-separable. Hence we can take finite sets $F_{n,k} \subset U_n \cap D_k$ $(k \geq n)$ such that every nonempty open set in U_n meets all but finitely many $F_{n,k}$. Let $F_n = \bigcup\{F_{k,n} : k \leq n\}$ for each $n \in \omega$. Then $F_n \subset D_n$ and every nonempty open set in $\bigcup\{U_n : n \in \omega\}$ meets all but finitely many F_n. Assume that the interior V of the set $\bigcap\{\overline{A \cap D_n} : n \in \omega\}$ is not empty. Then $V \cap A \cap D_n$ is dense in V, $V \cap A \cap D_n \subset V \cap A \subset V$ and $V \cap A$ is wH-separable. Hence there are finite sets $G_n \subset V \cap A \cap D_n$ $(n \in \omega)$ such that every nonempty open set in V meets all but finitely many G_n. Obviously $F_n \cup G_n \subset D_n$ and every nonempty open set in X meets all but finitely many $F_n \cup G_n$.

The case of wGN-separability follows from Lemma 2.7 (2). □

Corollary 2.9. *The finite union of H-separable spaces is wH-separable.*

Question 2.10. (1) Is there a wR-separable (resp., wH-separable, wGN-separable) space which is not R-separable (resp., H-separable, GN-separable)? (2) Is H-separability preserved under finite unions?

Proposition 2.11. *A space X is wR-separable if and only if for each sequence $\{D_n : n \in \omega\}$ of dense subsets of X, there are finite sets $F_n \subset D_n$ $(n \in \omega)$ such that $|F_n| \leq n$ and $\bigcup\{F_n : n \in \omega\}$ is dense in X.*

Proof. Assume that X is wR-separable, and let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X. Let $E_n = \bigcup\{D_k : k > n+1\}$ $(n \in \omega)$ and $E = \bigcap\{E_n : n \in \omega\}$. We put $V = X \setminus \overline{E}$ and $W = X \setminus \overline{V}$. Obviously $V \cup W$ is dense in X. Since V is wR-separable and each $E_n \cap V$ is dense in V, we can take points $x_n \in E_n \cap V$ $(n \in \omega)$ such that $\{x_n : n \in \omega\}$ is dense in V. Take some $\varphi \in \omega^\omega$ with $x_n \in D_{\varphi(n)}$ and $\varphi(n) > n+1$. For each $k \in \varphi(\omega)$, let $G_k = \{x_n : n \in \varphi^{-1}(k)\}$. Then $G_k \subset D_k, |G_k| \leq k-1$ and $\{x_n : n \in \omega\} = \bigcup\{G_k : k \in \omega\}$. Since W is wR-separable and $W \cap E$ is dense in W, we can take a subset $\{y_n : n \in \omega\} \subset W \cap E$ such that $\{y_n : n \in \omega\}$ is dense in W. Since each point of E is contained in D_n for infinitely many $n \in \omega$, there is a strictly increasing sequence $\{l_n : n \in \omega\} \subset \omega$ with $l_0 \geq 1$ and $y_n \in D_{l_n}$ $(n \in \omega)$. Now we put F_k $(k \in \omega)$ as follows:

$$F_k = \begin{cases} G_k \cup \{y_n\} & \text{if } k \in \varphi(\omega) \text{ and } k = l_n \\ G_k & \text{if } k \in \varphi(\omega) \text{ and } k \notin \{l_n : n \in \omega\} \\ \{y_n\} & \text{if } k = l_n \text{ and } k \notin \varphi(\omega) \\ \emptyset & \text{otherwise.} \end{cases}$$

Then $F_k \subset D_k, |F_k| \leq k$ and $\bigcup\{F_k : k \in \omega\}$ is dense in X. Conversely let $\{D_n : n \in \omega\}$ be a decreasing sequence of dense subsets of X. Let $E_n = D_{2^n}$ $(n \in \omega)$. Then there are subsets $F_n \subset E_n$ $(n \in \omega)$ such that $|F_n| \leq n$ and $\bigcup\{F_n : n \in \omega\}$ is dense in X. Obviously for each point $x \in \bigcup\{F_n : n \in \omega\}$, we can choose some $n_x \in \omega$ such that $x \in D_{n_x}$ and $n_x \neq n_y$ whenever $x \neq y$. □

We consider the algebraic sum of a selectively separable space. Regard $\mathbb{D}$ as the additive group $(\mathbb{Z}_2, +)$, where $\mathbb{Z}$ is the group of integers. Bella et al. asked in [6, Problem 3.11] whether the group generated by a selectively separable, countable and dense subset of $\mathbb{D}^\kappa$, where κ is some uncountable cardinal, is selectively separable. We show that there is a counterexample to this question. For the purpose, we recall the theorem below on selective separability of $C_p(X)$.

According to [21], a space X is said to be *Menger* if for every sequence $\{\mathcal{U}_n : n \in \omega\}$ of open covers of X, there are finite subfamilies $\mathcal{V}_n \subset \mathcal{U}_n$ $(n \in \omega)$ such that $\bigcup\{\mathcal{V}_n : n \in \omega\}$ is a cover of X. Every compact space is obviously Menger, and it is well-known that the space $\mathbb{P}$ of all irrationals is not Menger.

Theorem 2.12 ([5], Theorem 21, Proposition 25)**.** *For a Tychonoff space X, the following (1) and (2) are equivalent. Moreover, if X is zero-dimensional, (3) is also equivalent.*

(1) *$C_p(X)$ is selectively separable,*
(2) *X has a coarser second countable topology and each finite power of X is Menger,*
(3) *$C_p(X, \mathbb{D})$ is selectively separable.*

Example 2.13. Let X be a set with $|X| = \mathfrak{c}$. There is a selectively separable, countable and dense subset S in $\mathbb{D}^X$ such that neither the algebraic sum $S + S$ nor the group generated by S is selectively separable.

Proof. Let D_0 be any selectively separable, countable and dense subset in $\mathbb{D}^X$. For instance, since the Cantor set $\mathbb{C}$ satisfies the conditions in Theorem 2.12 (2), $C_p(\mathbb{C}, \mathbb{D})$ is selectively separable (moreover it is obviously countable and dense in $\mathbb{D}^X$), let $D_0 = C_p(\mathbb{C}, \mathbb{D})$. Let D_1 be any countable and dense subset in $\mathbb{D}^X$ which is not selectively separable. For instance, since the space $\mathbb{P}$ of all irrationals is not Menger, $C_p(\mathbb{P}, \mathbb{D})$ is not selectively separable by Theorem 2.12. Since $C_p(\mathbb{P}, \mathbb{D})$ is hereditarily separable (indeed it has a countable network), non-selective separability of $C_p(\mathbb{P}, \mathbb{D})$ implies that there is a countable and dense subset $D \subset C_p(\mathbb{P}, \mathbb{D})$ which is not selectively separable, let $D_1 = D$. For each $f \in \mathbb{D}^X$ and $A \subset X$, f_A is the element of $\mathbb{D}^X$ defined as follows: $f_A(x) = f(x)$ for any $x \in A$ and $f_A(x) = 0$ for any $x \in X \setminus A$. Let $\{X_0, X_1\}$ be any partition of X consisting of infinite subsets. We put

$$S = D_0 \cup \{f_{X_i} : i \in \{0, 1\}, f \in D_1\}.$$

Obviously S is countable and dense in $\mathbb{D}^X$. Since the set $\{f_{X_i} : i \in \{0, 1\}, f \in D_1\}$ is contained in the nowhere dense closed set $\{f_{X_i} : i \in \{0, 1\}, f \in \mathbb{D}^X\}$ in $\mathbb{D}^X$, the set $D_0 \setminus \{f_{X_i} : i \in \{0, 1\}, f \in \mathbb{D}^X\}$ is open and dense in S. Moreover $D_0 \setminus \{f_{X_i} : i \in \{0, 1\}, f \in \mathbb{D}^X\}$ is open in D_0, hence it is selectively separable by Lemma 1.2 (1). Thus S has a selectively separable, open and dense subset, therefore S is selectively separable by Lemma 1.2 (2). On the other hand, if $f \in D_1$, then $f = f_{X_0} + f_{X_1} \in S + S$. Thus $S + S$ has a dense subset which is not selectively separable, hence $S + S$ is not selectively separable. In particular, the group generated by S is not selectively separable. □

3. Products

L. Babinkostova [4, Corollary 2.5] showed that under CH, there are separable metric spaces X, Y such that $C_p(X)$ and $C_p(Y)$ are R-separable, but $C_p(X) \times C_p(Y)$

is not selectively separable. Under MA_{ctble}, Dow and Barman [8, Theorem 2.18] gave two countable selectively separable spaces whose product is not selectively separable.

In this section, under CH, we give two countable maximal R-separable spaces whose product is not selectively separable. In our next result, we will use the following lemma, part (i) of which is essentially Lemma 2.2 of [9].

Lemma 3.1. *Let (X, τ) be a countable regular dense-in-itself second countable space. (i) If A is dense in the topology τ, then there is a finer regular dense-in-itself second countable topology τ^* on X such that A is dense open in τ^*, and any dense open set in τ remains dense open in τ^*. (ii) If A is dense-in-itself in τ, then there is a finer regular dense-in-itself second countable topology τ^{**} on X such that either A is open in τ^{**}, or A has an isolated point in τ^{**}, and any dense open set in τ remains dense open in τ^{**}.*

Proof. Let $\{A_n : n \in \omega\}$ be a partition of A into dense sets, and let τ^* be the topology generated by

$$\tau \cup \{A_n : n \in \omega\} \cup \{X \setminus A_n : n \in \omega\}.$$

It is easy to check that this works for (i).

Let A be dense-in-itself in τ. If A is dense, we can apply (i). Otherwise, let $V = \mathrm{Int}_\tau A$. First add $\overline{V}$ to the topology τ. Since new open sets contain old ones, old dense open sets remain dense open. Then if $A \subset \overline{V}$, apply (i) to the dense set $A \cup (X \setminus \overline{V})$. This makes A open in the new topology. If on the other hand $A \not\subset \overline{V}$, choose $p \in A \setminus \overline{V}$, and apply (i) to the dense set

$$(X \setminus A) \cup \{p\} \cup \overline{V}.$$

This makes p isolated in A in the new topology. □

A space X is said to be *maximal* if it is dense-in-itself and no larger topology on X is dense-in-itself. If a space is dense-in-itself and every dense-in-itself subset is open, then it is maximal: see van Douwen [7, Theorem 2.2].

Example 3.2. Assuming CH, there is a countable regular maximal space X which is R-separable but X^2 is not selectively separable.[1]

Proof. We follow roughly the construction of Example 3.4 in [11] giving a countable regular space X such that X^2 has no slim dense set. Here, $D \subset X^2$ is called *slim* if D meets every cross-section $C^i_p = \{(x_0, x_1) \in X^2 : x_i = p\}$, where $p \in X$ and $i < 2$, in a set which is nowhere-dense in C^i_p.

Let us see first that if X is countable dense-in-itself and X^2 has no slim dense set, then X^2 is not selectively separable. Let $X = \{x_n : n \in \omega\}$, and let $D_n = \{(x, y) : x, y \notin \{x_i : i \leq n\}\}$. Clearly each D_n is dense in X^2, and if F_n is a finite subset of D_n, then $\bigcup\{F_n : n \in \omega\}$ meets each cross-section in a finite, hence nowhere-dense, set, so cannot be dense in X^2.

We now begin our construction of X. The set for X is ω. Let τ_0 be any regular dense-in-itself second countable topology on X. List all infinite subsets of ω in the sequence $\{A_\alpha : \alpha < \omega_1\}$, all subsets of ω^2 in the sequence $\{E_\alpha : \alpha < \omega_1\}$, and all ω-sequences of subsets of ω in the sequence $\{\langle D^\alpha_n \rangle_{n \in \omega} : \alpha < \omega_1\}$, such that each

[1] A. Bella and the first author have observed that this example can also be constructed assuming MA_{ctble}.

set appear in these sequences ω_1 times. For $\alpha < \omega_1$ we will construct a regular dense-in-itself second countable topology τ_α on ω such that

(1) If $\beta < \alpha$, then $\tau_\beta \subset \tau_\alpha$;
(2) If $\beta < \alpha$ and U is dense open in τ_β, then U is dense open in τ_α;
(3) If A_α is dense in τ_α, then A_α is dense open in $\tau_{\alpha+1}$;
(4) If A_α is not dense in τ_α, but is dense-in-itself, then either A_α is open in $\tau_{\alpha+1}$ or A_α has an isolated point in $\tau_{\alpha+1}$;
(5) If D_n^α, $n \in \omega$, are dense open sets in τ_α, then there are $x_n^\alpha \in D_n^\alpha$ $(n \in \omega)$ such that $F^\alpha = \{x_n^\alpha : n \in \omega\}$ is dense open in $\tau_{\alpha+1}$;
(6) E_α is not slim dense in $\tau_{\alpha+1}^2$.

Suppose we have constructed τ_β and x_n^β satisfying the above conditions for all $\beta < \alpha$, where $\alpha < \omega_1$.

If α is a limit ordinal, we let $\tau_\alpha = \bigcup\{\tau_\beta : \beta < \alpha\}$, and there is nothing else to do. Now suppose $\alpha = \gamma + 1$. Look at A_γ. If A_γ is dense in τ_γ, use part (i) of the lemma to obtain a finer regular dense-in-itself second countable topology τ'_γ such that A_γ is dense open in τ'_γ, and any dense open set in τ_γ remains dense open in τ'_γ. If A_γ is not dense in τ_γ but is dense-in-itself, apply instead part (ii) of the lemma to obtain τ'_γ satisfying the conclusion of (ii) with $A = A_\gamma$.

Next, if each D_n^γ is dense open in τ_γ, then it is dense open in τ'_γ, so we can choose points $x_n^\gamma \in D_n^\gamma$ $(n \in \omega)$ such that $F^\gamma = \{x_n^\gamma : n \in \omega\}$ is dense in τ'_γ. Then apply part (i) of the lemma again to obtain a finer topology τ''_γ in which F^γ is dense open.

Finally, suppose E_γ is slim dense in $\tau''_\gamma \times \tau''_\gamma$. Let $\{U_n \colon n < \omega\}$ be a sequence of basic open sets of τ''_γ. Choose inductively a sequences a_n, $n < \omega$, of points of X such that

(i) $a_n \in U_n \setminus \{a_i \colon i < n\}$;
(ii) $(\{a_i \colon i \le n\}^2 \setminus \Delta_X) \cap E_\gamma = \emptyset$, where $\Delta_X = \{(x,x) : x \in X\}$.

Such a choice is possible, because the set

$$H_n = \{x \in X \colon \text{for some } i < n, (x, a_i) \in E_\gamma \text{ or } (a_i, x) \in E_\gamma\}$$

is nowhere dense (as E_γ is slim), so $U_n \setminus H_n$ is infinite. Let $T_\alpha = \{a_n \colon n < \omega\}$. Then T_α is dense in τ''_γ and $(T_\alpha^2 \setminus \Delta_X) \cap E_\gamma = \emptyset$. Apply part (i) of the lemma yet again to obtain the topology τ_α in which T_α is dense open. Clearly, E_γ is not dense in τ_α^2. This completes the inductive construction.

Let $\tau = \bigcup\{\tau_\alpha : \alpha < \omega_1\}$. We check that the topology τ on X satisfies the conditions of the theorem. Suppose A is dense-in-itself in τ. Then A is dense-in-itself in every prior topology τ_α, $\alpha < \omega_1$, so at some point, A was made either open or made to have an isolated point. Since the latter can't be the case, A is open in τ. Thus (X, τ) is maximal.

Suppose D_n, $n \in \omega$, is a sequence of τ-dense sets. Then D_n is dense in every prior topology, so for some $\alpha < \omega_1$, $D_n = D_n^\alpha$ and D_n^α is dense open in τ_α for all n. Then by the construction, there are $x_n^\alpha \in D_n^\alpha$ such that $F^\alpha = \{x_n^\alpha : n \in \omega\}$ is dense open in $\tau_{\alpha+1}$, hence dense open in τ. So (X, τ) is R-separable.

Finally we will verify that there is no dense slim set in X^2, hence X^2 is not selectively separable. Suppose E is such. Since X is countable, every open set U in X (resp., X^2) is the union of countably many basic open sets, and hence is open in (X, τ_α) (resp., $(X, \tau_\alpha)^2$) for some $\alpha < \omega_1$. Since τ_α is a weaker topology, if U is also dense in X, it is dense in the weaker topology. It follows that, since there are

only countably many cross-sections to consider, E is also a slim dense set in some (X, τ_α). Let $\beta \geq \alpha$ such that $E_\beta = E$. Then by the construction, E_β is not dense in $(X, \tau_{\beta+1})^2$, contradiction. □

The existence of a maximal selectively separable space is independent of ZFC: see [8].

4. MISCELLANEOUS RESULTS

According to [16], a space X is said to be a *Pytkeev space* if $x \in \overline{A} \setminus A$ and $A \subset X$ imply the existence of a family $\{A_n : n \in \omega\}$ of infinite subsets of A such that every neighborhood of x contains some A_n. Note that the family $\{A_n : n \in \omega\}$ can be pairwise-disjoint. This notion was originally introduced in Pytkeev [19], and every subsequential space is a Pytkeev space [19], where a space is said to be *subsequential* if it is homeomorphic to some subspace of a sequential space.

Proposition 4.1. *Let X be a Pytkeev space. If X is selectively separable, then it is R-separable. Moreover, if X is dense-in-itself, then it is GN-separable.*

Proof. Let $\{D_n : n \in \omega\}$ be a sequence of dense subsets of X and let $\{x_n : n \in \omega\}$ be a dense subset of X. We put $\{D_n : n \in \omega\} = \{D_{n,m} : n, m \in \omega\}$. For each $n, m \in \omega$, we can take finite subsets $F_{n,m} \subset D_{n,m}$ $(n, m \in \omega)$ such that each $G_n = \bigcup\{F_{n,m} : m \in \omega\}$ is dense in X. We put $I = \{n \in \omega : x_n \in G_n\}$ and $J = X \setminus I$. If $n \in I$, then $x_n \in D_{n,m}$ for some $m \in \omega$. If $n \in J$, then using Pytkeev's property, we can take a family $\{A_{n,m} : m \in \omega\}$ of infinite subsets of G_n such that every neighborhood of x_n contains some $A_{n,m}$. Note that $A_{n,m} \cap F_{n,k} \neq \emptyset$ holds for infinitely many $k \in \omega$. Hence there are a strictly increasing sequence $\{k_m : m \in \omega\} \subset \omega$ and points $y_{n,k_m} \in A_{n,m} \cap F_{n,k_m} \subset D_{n,k_m}$ $(m \in \omega)$. Obviously $x_n \in \overline{\{y_{n,k_m} : m \in \omega\}}$. Thus $\{x_n : n \in I\} \cup \{y_{n,k_m} : n \in J, m \in \omega\}$ is dense in X. This shows that X is R-separable.

Assume that X is selectively separable, dense-in-itself and Pytkeev. To show that X is GN-separable, we have only to show that X is wH-separable by Lemma 2.7. Let $\{D_n : n \in \omega\}$ be a decreasing sequence of dense subsets of X. Since X is R-separable by the former part of this lemma, there are points $x_n \in D_n$ $(n \in \omega)$ such that the set $D = \{x_n : n \in \omega\}$ is dense in X. Since D is dense-in-itself, $x_n \in \overline{D \setminus \{x_n\}}$, hence for each $n \in \omega$ there are infinite subsets $A_{n,m} \subset D \setminus \{x_n\}$ $(m \in \omega)$ such that every open neighborhood of x_n contains some $A_{n,m}$. We enumerate $\{A_{n,m} : n, m \in \omega\} = \{B_n : n \in \omega\}$. Since $\{D_n : n \in \omega\}$ is decreasing and each B_n is infinite, $B_n \cap D_m \neq \emptyset$ for all $n, m \in \omega$. Therefore, for each $n \in \omega$ and $0 \leq j \leq n$ we can take a point $y_{n,j} \in D_n \cap B_j$. Let $F_n = \{y_{n,j} : 0 \leq j \leq n\}$ $(n \in \omega)$. Then each F_n is a finite subset of D_n. Let U be a nonempty open subset of X. Since D is dense in X, $x_n \in U$ for some $n \in \omega$, hence $A_{n,m} \subset U$ for some $m \in \omega$. Let $A_{n,m} = B_k$. Then $y_{l,k} \in D_l \cap B_k \subset U$ $(l \geq k)$, thus $F_l \cap U \neq \emptyset$ $(l \geq k)$. □

Dow and Barman [8] proved that each separable Fréchet space is selectively separable. The following is an improvement of Dow and Barman's result.

Corollary 4.2. *Each separable Fréchet space X is R-separable. Moreover, if X is dense-in-itself, then it is GN-separable.*

Bella et al. asked in [6, Problem 3.5] whether every separable Čech-complete space with countable tightness is selectively separable. Recall that every Čech-complete space is of pointwise countable type. Šapirovskiĭ [20, Theorem 1″] showed

that if a space X with countable tightness is of pointwise countable type, then π-character of X is countable. Hence every separable Čech-complete space with countable tightness has a countable π-base (in particular selectively separable). Generally, even if a space X is separable, has countable tightness and each point $x \in X$ is contained a countably compact subset $K_x \subset X$ such that $\chi(K_x, X) \leq \omega$, X need not have a countable π-base. Indeed, under CH, Hajnal and Juhász [12] constructed a countably compact hereditarily separable space with uncountable π-character. However we can show that if a space is separable, has countable tightness and each point $x \in X$ is contained in a countably compact subset $K_x \subset X$ such that $\chi(K_x, X) \leq \omega$, then it is selectively separable.

Lemma 4.3. *Let X be a space. Suppose that for each point $x \in X$, there is a subset $K_x \subset X$ such that $x \in K_x$, $\chi(K_x, X) \leq \omega$ and K_x has countable fan-tightness, then X has countable fan-tightness.*

Proof. Assume $x \in \bigcap\{\overline{A_n} : n \in \omega\}$, where $A_n \subset X$. Take a subset $K_x \subset X$ such that $x \in K_x$, $\chi(K_x, X) \leq \omega$ and K_x has countable fan-tightness. If there is a strictly increasing sequence $\{n_j : n \in \omega\} \subset \omega$ such that $x \in \bigcap\{\overline{K_x \cap A_{n_j}} : j \in \omega\}$, by countable fan-tightness of K_x we can take finite sets $F_{n_j} \subset K_x \cap A_{n_j}$ $(j \in \omega)$ such that $x \in \overline{\bigcup\{F_{n_j} : j \in \omega\}}$. Therefore, without loss of generality we may assume $K_x \cap A_n = \emptyset$ for each $n \in \omega$. For each $\varphi = (a_n) \in \mathbb{A} = \prod\{A_n : n \in \omega\}$, let $F_\varphi = K_x \cap \overline{\{a_n : n \in \omega\}}$. Assume $x \in X \setminus \overline{\bigcup\{F_\varphi : \varphi \in \mathbb{A}\}}$, and take an open neighborhood V of x with $\overline{V} \cap \overline{\bigcup\{F_\varphi : \varphi \in \mathbb{A}\}} = \emptyset$. Let $\{U_n : n \in \omega\}$ be an open neighborhood base of K_x in X. For each $n \in \omega$, take a point $b_n \in V \cap U_n \cap A_n$, and let $\psi = (b_n)$. Then $F_\psi \neq \emptyset$ and $F_\psi \subset \overline{V}$. This is a contradiction. Hence $x \in \overline{\bigcup\{F_\varphi : \varphi \in \mathbb{A}\}}$. Since the tightness of K_x is countable, there is a subset $\{x_n : n \in \omega\} \subset \bigcup\{F_\varphi : \varphi \in \mathbb{A}\}$ such that $x \in \overline{\{x_n : n \in \omega\}}$. Let $x_n \in F_{\varphi_n}$ and $\varphi_n = (a_{n,m}) \in \mathbb{A}$. Then $x_n \in \overline{\{a_{n,m} : m \geq n\}}$. Hence if we put $F_n = \{a_{k,n} : k \leq n\}$ for each $n \in \omega$, we have $F_n \subset A_n$ and $x \in \overline{\bigcup\{F_n : n \in \omega\}}$. Thus X has countable fan-tightness. □

Theorem 4.4. *Let X be a separable space with countable tightness. If each point $x \in X$ is contained in a countably compact subset $K_x \subset X$ such that $\chi(K_x, X) \leq \omega$, then X is selectively separable.*

Proof. Arhangel'slii and Bella [3, Corollary 2] showed that every countably compact space with countable tightness has countable fan-tightness. Hence each K_x has countable fan-tightness. By Lemma 1.2 (3) and Lemma 4.3, X is selectively separable. □

We conclude this paper with a remark on strong density. According to [23], for a space X we put $sd(X) = \sup\{d(D) : D$ is dense in $X\}$, where $d(D)$ is the density of D. Weston and Shilleto [23] gave a Hausdorff (non-regular) space X with $sd(X) = \omega$ and $sd(X \times X) > \omega$, and asked whether there is such a regular example.

Example 4.5. There is a Tychonoff space Z with $sd(Z) = \omega$ and $sd(Z \times Z) > \omega$.

Proof. Okunev and Tkachuk [18, Corollary 2.9] showed that $sd(C_p(X)) = \omega$ if and only if $C_p(X)$ is separable and has countable tightness. It is known [2] that $C_p(X)$ is separable if and only if X has a coarser second countable topology, and $C_p(X)$

has countable tightness if and only if each finite power of X is Lindelöf. Hence $sd(C_p(X)) = \omega$ if and only if X has a coarser second countable topology and each finite power of X is Lindelöf. According to [14], for a Bernstein set A in the real line $\mathbb{R}$ we denote by $\mathbb{L}(\mathbb{R}, A)$ the Lindelöf refinement of $\mathbb{R}$ obtained by isolating all points of $\mathbb{R} \setminus A$. Lawrence [14] showed that there is a Bernstein set $A \subset \mathbb{R}$ such that each finite power of $\mathbb{L}(\mathbb{R}, A)$ is Lindelöf. On the other hand, using Michael's method [17] $\mathbb{L}(\mathbb{R}, A) \times (\mathbb{R} \setminus A)$ is not normal (hence not Lindelöf). Therefore $sd(C_p(\mathbb{L}(\mathbb{R}, A))) = sd(C_p(\mathbb{R} \setminus A)) = \omega$, but $sd(C_p(\mathbb{L}(\mathbb{R}, A)) \times C_p(\mathbb{R} \setminus A)) = sd(C_p(\mathbb{L}(\mathbb{R}, A) \oplus \mathbb{R} \setminus A)) > \omega$. If we put $Z = C_p(\mathbb{L}(\mathbb{R}, A)) \oplus C_p(\mathbb{R} \setminus A)$, then it is a Tychonoff space with $sd(Z) = \omega$ and $sd(Z \times Z) > \omega$. □

Acknowledgement: The authors thank Alan Dow for information and comments concerning this paper.

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